The German International University for Applied Sciences

Faculty of Business Informatics

Digital Transformation

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Project PD1

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1. Company Overview and Heritage

Year of Establishment:
Arab Contractors was established in 1955

Industry/Sector:
The company operates in the construction and engineering sector, delivering infrastructure, housing, transportation, and large scale national projects.

Founders and Key Figures:
The company was founded by Osman Ahmed Osman, one of Egypt's most influential engineers and former Minister of Housing.

Historical Significance:
Arab Contractors is one of the largest state owned construction firms in the Middle East and Africa, with over 70,000 workers and operating in more than 29 countries.

Key milestones and achievements before digital transformation:

- Participation in major Egyptian national projects such as:
- Roads and bridges development programs
- Infrastructure projects linked to the Suez Canal region
- Expansion into African markets including Nigeria, Uganda, and Algeria
- Strong partnerships with governments for public infrastructure

Traditional business model and operational processes:

- Manual, paper based documentation systems
- Decentralized project management across sites
- Limited use of integrated software systems
- Heavy reliance on human expertise rather than data-driven decision-making

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2. The Need for Digital Transformation

The Requirement for Digital Change

Marketplace & Industry Background

Rapid transformations in the worldwide construction sector because of digital disruption. Examples are:

• The use of Building Information Modeling (BIM) and intelligent construction technology

• Multinational organizations are now competing for business using next generation tools

• Clients increasingly demand delivery that is faster, less expensive and more real-time transparent

• The Egyptian government is promoting digital infrastructure / smart cities

In Egypt, the efficiencies and coordination of large-scale national projects (e.g., New Administrative Capital) have forced companies like Arab Contractors to have to innovate their operations.

Examples of Major Challenges Experienced Prior to Digital Transformation

• Business processes that were outdated. i.e., companies used paper-based methods to create workflows which causes delays and errors.

• The supply chain was inefficient as real-time data regarding the location of materials was not available, contributing to materials shortages and cost overruns.

• Manual processing of the majority of operations led to lower productivity and slower reporting cycles.

• Poor communication between departments as a result of data silos existed.

• Increased competition from international firms that have adigital technology resulted in the ability to deliver construction projects with faster lead times at a lower cost.

3. Initiatives Towards Digital Transformation

Strategic Vision and Leadership:

The Company's leadership have recognized that continuing with their old legacy systems would limit their future growth, as a result they began focusing on a more digital approach in order to:

- Reduce manual work
- Improve operational efficiency
- Support their long term goals
- Improve decision making using data

One of the most important drivers behind this shift is the reduction of paper usage in order to move towards a more modern and environmentally responsible organization.

Adopted Technologies and Digital Initiatives:

The Arab Contractors have implemented several digital solutions such as:

1. **ERP Systems**

The Arab Contractors have started to implement Enterprise Resource Planning systems in order to connect different parts of the organization.

These systems help in integrating Financial Operations, inventory management, project coordination and procurement.

ERP allows for everything to be connected under one system, instead of each department working independently, which as a result it improves accuracy, coordination and decision making.

2. **Document Cloud Systems and Cloud Tools**

The Arab Contractors have moved towards digital document management, in order to reduce paper consumption.

The old days of storing and printing documents are over, now company files are stored digitally, accessed remotely and shared between departments.

Cloud based tools also help employees working efficiency and reduce the delays caused by manual approvals.

3. **Data Analytics Tools (Power BI)**

In order to get a better understanding of its operations, The Arab Contractors starting using data analytics tools, especially Microsoft Power BI.

The data usually comes in the form of excel sheets and applications like Power BI help generate dashboards that are used to track costs and operational performance.

This allows the company to identify problems much quicker and make decisions based on actual data.

4. **Digital Workflow automation**

The Arab Contractors have moved from manual processes to automated digital workflows.

Instead of manually printing documents, sending them manually and waiting for approval. The company can now automate approvals, send notifications, and track progress.

As a result, this saves time, reduce errors and improve coordination between departments.

5. **AI and Advanced Technologies (For the Future)**

The company is thinking long term and are looking to adopt more advanced technologies like AI and Smart project tracking systems that would predict delays, manage risks and improve the projects performance.

Impact on Internal Operations and Workforce:

Organizational Changes: to manage this digital transformation, The Arab Contractors are looking to create a more structured system, which will include teams responsible for digital initiatives to ensure the transformation is not random and ensure its alignment with the company's goals.

Employee Training: One of the challenges the Arab Contractors is facing, is that their employees are not familiar with the new systems introduced. In order to make their employees feel comfortable using the new systems, they have focused on training programs, workshops and continuous support to their employees.

Resistance to Change: Resistance to Change is completely normal in any big organization. In the Arab Contractors employees prefer paper

documents and traditional workflow, this mainly due to old habits, lack of training and fear of change. In order to overcome this the company is gradually introducing systems while encouraging its employees to adapt.

4. Current Situation and Observations

Benefits:

- **Improved operational efficiency:**
  Automation reduced project delays and manual errors
- **Enhanced customer engagement:**
  Better transparency and communication with stakeholders
- **Cost reduction:**
  Optimized resource allocation and minimized waste
- **Improved decision-making:**
  Real time data enabled faster and more accurate decisions
- **Stronger competitive position:**
  Ability to compete with international construction firms

Challenges Encountered:

- **Resistance to change:**
  Some employees struggled to adapt to new technologies
- **Technical integration issues:**
  Difficulty integrating legacy systems with new platforms
- **High financial investment:**
  High costs needed a lot of infrastructure and training

Future Prospects

- Increased adoption of AI and automation in construction

- Expansion of smart construction and IoT technologies
- Greater reliance on data analytics and predictive modeling
- Continued alignment with Egypt's Vision 2030 digital transformation strategy

These initiatives are expected to ensure long-term sustainability and global competitiveness.

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